Data Flow Diagrams (DFD) & Data Dictionary

Academic Grievance, No-Dues & Clearance Automation Engine

Standard: Structured Systems Analysis & Design (SSAD) — **Yourdon & DeMarco / Gane & Sarson Notation**
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Table of Contents

1. Structured Systems Analysis (SSAD) Notational Guide & Formal Rules
    - 1.1 Notational Conventions & Symbol Guide
    - 1.2 Formal DFD Construction Rules & Quality Compliance
2. DFD Level 0: Context Analysis Diagram
    - 2.1 Environmental Boundary & Entity Characterization
    - 2.2 Level 0 Context Diagram (Mermaid Model)
    - 2.3 Context-Level Input and Output Boundary Data Flows
3. DFD Level 1: System Decomposition Diagram
    - 3.1 Functional Decomposition Strategy (Processes 1.0 through 6.0)
    - 3.2 Persistent Data Stores Overview (`DS–1` through `DS–6`)
    - 3.3 Level 1 System Decomposition Diagram (Mermaid Model)
    - 3.4 Level 1 Inter-Process, Store, and Entity Data Flows
4. Process Specifications (P-Specs), Decision Tables & Decision Trees
    - 4.1 Process 3.0: Chain of Responsibility (CoR) Gating & Multi-Signatory Barrier Logic
        - Decision Table 1: Multi-Gate Clearance Pipeline & Prerequisite Barrier Matrix
        - Decision Tree 1: Clearance Gating & Barrier Enforcement Tree (Mermaid)
        - Decision Tree 2: Departmental Liability Assessment & HMAC Signing Flow (Mermaid)
    - 4.2 Process 4.0: Grievance Redressal & SLA Escalation Engine
        - Decision Table 2: Priority SLA Deadline & Automated Escalation Matrix
        - Decision Tree 3: Grievance Arbitration & SLA Breach Escalation Tree (Mermaid)
    - 4.3 Process 1.0 & 3.0: Role-Based Access Control (RBAC) & Authority Validation
        - Decision Table 3: Comprehensive RBAC Authorization Matrix
        - Decision Tree 4: User Authentication & Authorization Gate Tree (Mermaid)
5. Comprehensive Data Dictionary
    - 5.1 Data Stores at Rest (`DS–1` to `DS–6` Schemas)
    - 5.2 Data Elements in Motion (Composite Packets & Primitive Elements)
6. Formal Horizontal & Vertical Balancing Verification

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1. Structured Systems Analysis (SSAD) Notational Guide & Formal Rules

1.1 Notational Conventions & Symbol Guide

This specification strictly follows the standardized graphical symbols of **Gane & Sarson** and **Yourdon & DeMarco**:

Symbol Element	Gane & Sarson / Yourdon Symbol	Mermaid Semantic Representation	Architectural Definition
External Entity	Double Square / Rectangle	`Entity["External Entity Name"]`	An autonomous origin or destination of data residing outside system boundaries (e.g., Student, Staff).
Process	Rounded Box / Circle	`Process(("ID: Process Name"))`	An algorithmic transformation converting incoming data packets into outgoing information flows.
Data Store	Open-Ended Rectangle	`Store[("DS-ID: Data Store Name")]`	A persistent repository holding system data at rest (e.g., relational database tables).
Data Flow	Directed Vector Arrow	`A -- "Data Flow Name" --> B`	A deterministic pipeline conveying structured data packets between entities, processes, and stores.

1.2 Formal DFD Construction Rules & Quality Compliance

To ensure mathematical consistency, the diagrams conform to the classical **Six Golden Rules of Structured Analysis**:

```
+-----------------------------------------------------------------------------------------------+
|                            SIX GOLDEN RULES OF STRUCTURED DFD DESIGN                          |
+-----------------------------------+-----------------------------------------------------------+
| Rule 1: Conservation of Data      | No "Miracles" (output without input) or "Black Holes"
|                                   | (input without output). Every process transforms data.
+-----------------------------------+-----------------------------------------------------------+
| Rule 2: Entity-to-Entity Prohibited| External entities cannot communicate directly inside the DFD
|                                   | boundary; all interaction is mediated by a process.
+-----------------------------------+-----------------------------------------------------------+
| Rule 3: Store-to-Store Prohibited | Data stores cannot transfer data directly between themselves
|                                   | without an executing process reading and writing.
+-----------------------------------+-----------------------------------------------------------+
| Rule 4: Entity-to-Store Prohibited| External entities cannot directly read or write to internal
|                                   | tables without an intervening processing step.
+-----------------------------------+-----------------------------------------------------------+
| Rule 5: Horizontal & Vertical Parity| Data flows entering/leaving Level 0 Context must balance
|                                   | perfectly with the aggregate boundary flows at Level 1.
+-----------------------------------+-----------------------------------------------------------+
| Rule 6: Decimal Numbering Hierarchy| Unique decimal indexing: Context is 0.0, Level 1 processes
|                                   | are 1.0, 2.0, 3.0, 4.0, 5.0, 6.0; Data stores are DS-1..DS-6.
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```

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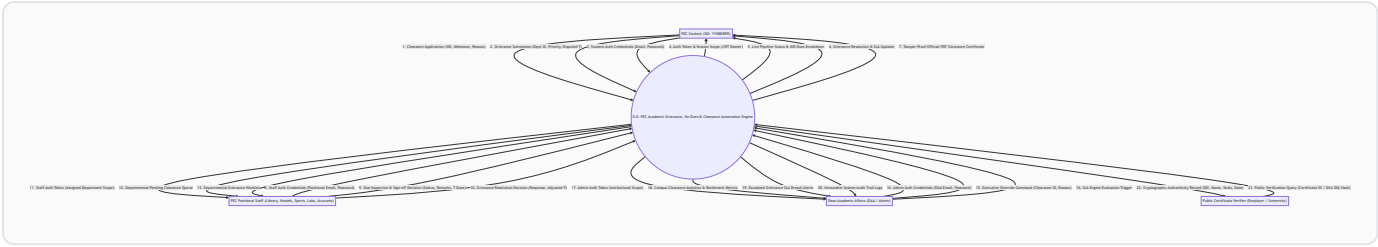
2. DFD Level 0: Context Analysis Diagram

2.1 Environmental Boundary & Entity Characterization

The Context Diagram defines the outermost functional boundary of the **PEC Academic Grievance, No-Dues & Clearance Automation Engine** within **Punjab Engineering College (Sector 12, Chandigarh)**. The central engine (`0.0`) mediates all transactions between four primary external entities:

1. **PEC Student (`Student`):**
   - Identified by an immutable 8-digit PEC SID (`YYBBBRRR` , e.g., `22103042`).
   - Initiates clearance applications, monitors real-time gate progress, views itemized dues in Indian Rupees (`₹`), files grievance appeals, and downloads official tamper-evident PDF certificates.
2. **Positional Department Staff (`Staff`):**
   - Permanent administrative designations: *Librarian* (Dr. B.R. Ambedkar Central Library), *Chief Warden* (PEC Hostels), *President Gymkhana* (Sports Complex), *HOD Labs* (Department Computing & Research Labs), and *Accounts Officer* (Caution Money & Finance Branch).
   - Inspects asset return records, records financial liabilities in INR (`₹`), grants cryptographic sign-offs, and resolves student grievances.
3. **Dean Academic Affairs (`Admin` / DAA):**
   - Chief institutional academic authority overseeing campus-wide clearance compliance.
   - Monitors clearance queues, departmental bottlenecks, and SLA breach escalations; applies executive overrides with mandatory audit justification.
4. **Public Certificate Verifier (`Verifier`):**
   - External employers, background verification agencies, and foreign universities.
   - Validates clearance certificates tokenlessly using Certificate ID or SHA-256 hash digests.

2.2 Level 0 Context Diagram (Mermaid Model)



2.3 Context-Level Input and Output Boundary Data Flows

Flow Index	Flow Name	Direction	Source	Destination	Data Composition / Packet Description
F-01	`Clearance Application`	Input	`Student`	Process `0.0`	`student_sid` + `academic_year` + `semester` + `reason`
F-02	`Grievance Submission`	Input	`Student`	Process `0.0`	`student_id` + `department_id` + `subject` + `description` + `priority`
F-03	`Student Auth Credentials`	Input	`Student`	Process `0.0`	`email` (`@pec.edu.in`) + `password`
F-04	`Auth Token & Session Scope`	Output	Process `0.0`	`Student`	`access_token` (JWT) + `user_id` + `role: STUDENT` + `roll_number`
F-05	`Live Pipeline Status`	Output	Process `0.0`	`Student`	`overall_status` + 5 $\times$ (`dept_name` , `gate` , `status` , `₹ due_amount` , `remarks`)
F-06	`Grievance Resolution Updates`	Output	Process `0.0`	`Student`	`grievance_id` + `status` + `staff_response` + `resolved_at`
F-07	`Official PDF Certificate`	Output	Process `0.0`	`Student`	Binary vector PDF (`%PDF-`) + DAA/DSA letterhead + SHA-256 seal
F-08	`Staff Auth Credentials`	Input	`Staff`	Process `0.0`	Positional email (`library@pec.edu.in` , etc.) + `password`
F-09	`Due Inspection & Sign-off`	Input	`Staff`	Process `0.0`	`approval_id` + `decision` (`APPROVED` / `REJECTED`) + `remarks` + `₹ due_amount`
F-10	`Grievance Resolution Decision`	Input	`Staff`	Process `0.0`	`grievance_id` + `staff_response` + `status: RESOLVED`
F-11	`Staff Auth Token`	Output	Process `0.0`	`Staff`	`access_token` (JWT) + `role: STAFF` + `department_id`
F-12	`Department Review Queue`	Output	Process `0.0`	`Staff`	List of pending approvals filtered strictly by staff's `department_id`
F-13	`Department Grievance Worklist`	Output	Process `0.0`	`Staff`	List of grievances filtered by `department_id` with active SLA countdowns

Flow Index	Flow Name	Direction	Source	Destination	Data Composition / Packet Description
F-14	`Admin Auth Credentials`	Input	`Admin`	Process `0.0`	`admin@pec.edu.in` + `password`
F-15	`Executive Override Command`	Input	`Admin`	Process `0.0`	`clearance_id` + `department_code` + `reason` (`ADMIN_OVERRIDE`)
F-16	`SLA Evaluation Trigger`	Input	`Admin`	Process `0.0`	Trigger signal for real-time background SLA deadline evaluation
F-17	`Admin Auth Token`	Output	Process `0.0`	`Admin`	`access_token` (JWT) + `role: ADMIN`
F-18	`Campus Clearance Analytics`	Output	Process `0.0`	`Admin`	Total counts (`approved` , `pending` , `rejected`) + Dept bottleneck hours
F-19	`Escalated Grievance Alerts`	Output	Process `0.0`	`Admin`	List of breached tickets with `is_escalated = True` for DAA action
F-20	`System Audit Trail Logs`	Output	Process `0.0`	`Admin`	Immutable append-only log entries (`actor` , `action` , `target` , `timestamp`)
F-21	`Public Verification Query`	Input	`Verifier`	Process `0.0`	`cert_id` (`PEC-CLR-YYYY-XXXXX`) OR 64-character SHA-256 digest
F-22	`Authenticity Record`	Output	Process `0.0`	`Verifier`	`valid: Boolean` + `student_name` + `sid` + `session` + 5 $\times$ HMAC Seals

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3. DFD Level 1: System Decomposition Diagram

3.1 Functional Decomposition Strategy (Processes 1.0 through 6.0)

The Level 1 DFD decomposes the top-level process `0.0` into six dedicated, single-responsibility operational subsystems:

- **`1.0: User Authentication & Role Authorization`** : Validates user credentials, generates Bcrypt hashes, issues signed JWT access tokens, and enforces Role-Based Access Control (RBAC).
- **`2.0: Clearance Application & Workflow Initialization`** : Enforces the single-active-request rule, captures student academic milestones, and initializes the 5 departmental approval records across the 3 gates.

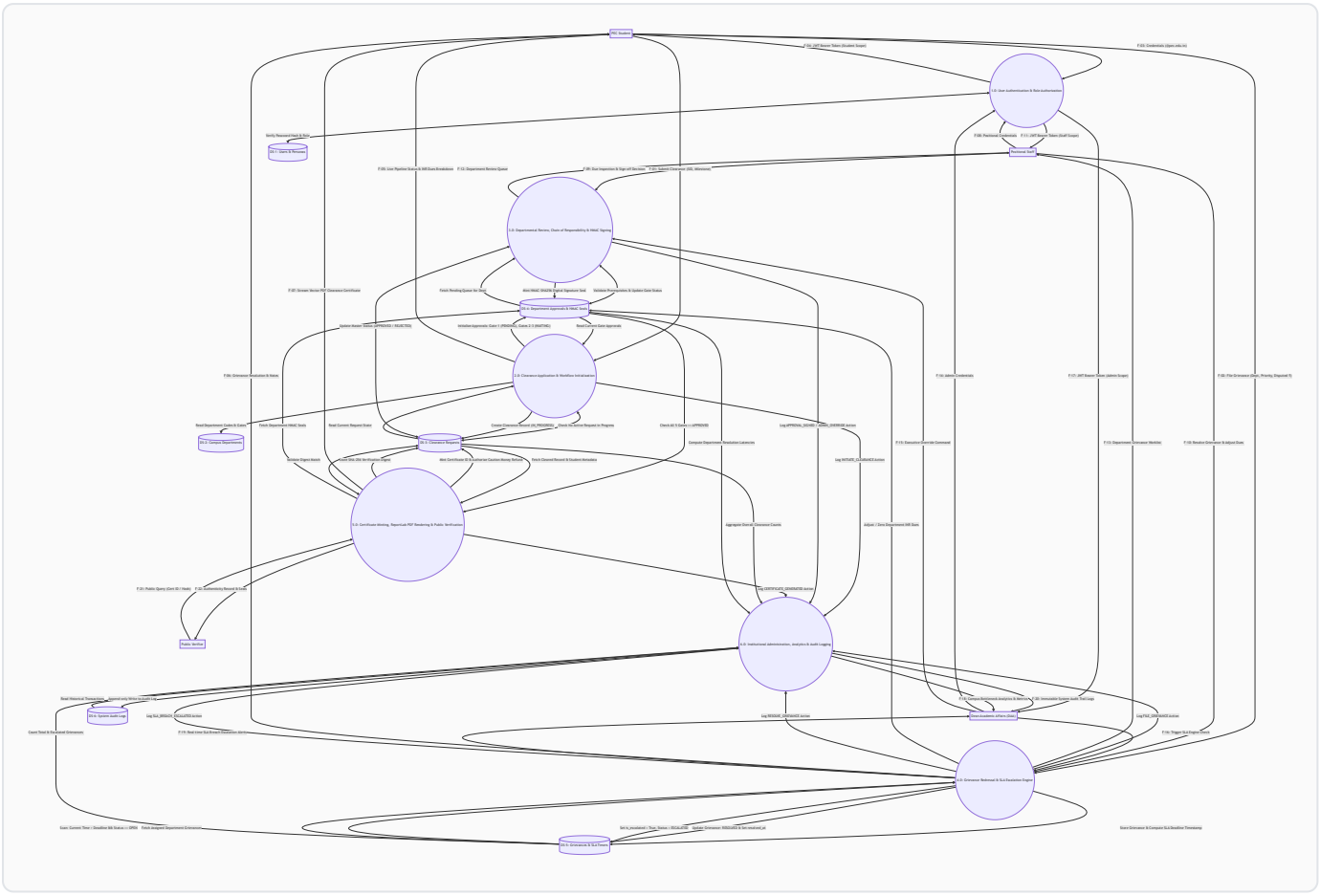
- **`3.0: Departmental Review, Chain of Responsibility & HMAC Signing`:** Orchestrates the multi-department clearance pipeline, allows parallel review at Gate 1, enforces prerequisite barriers at Gate 2 and Gate 3, generates HMAC-SHA256 digital seals, and executes DAA overrides.
- **`4.0: Grievance Redressal & SLA Escalation Engine`:** Manages student disputes, tags priority SLA turnaround windows (4h, 12h, 24h, 48h), evaluates deadline breaches, and auto-escalates delinquent tickets to the DAA.
- **`5.0: Certificate Minting, ReportLab PDF Rendering & Public Verification`:** Mints unique certificate identifiers upon Gate 3 completion, releases the ₹10,000.00 Caution Money deposit, computes SHA-256 verification digests, renders vector PDF certificates, and serves tokenless public authenticity verifications.
- **`6.0: Institutional Administration, Analytics & Audit Logging`:** Computes facility bottleneck statistics, provides institutional oversight for the DAA, and immutably persists all lifecycle transactions into an append-only audit trail.

3.2 Persistent Data Stores Overview (`DS-1` through `DS-6`)

The system state is maintained across six relational data stores:

- **`DS-1: Users & Personas`** (**`users`** table): User identities, authentication hashes, roles (`STUDENT`, `STAFF`, `ADMIN`), and department linkages.
- **`DS-2: Campus Departments`** (**`departments`** table): Academic facilities, institutional codes, gate assignments, and staff head designations.
- **`DS-3: Clearance Requests`** (**`clearance_requests`** table): Master student clearance records, milestone metadata, overall statuses, and certificate digests.
- **`DS-4: Department Approvals & HMAC Seals`** (**`department_approvals`** table): Departmental review records, itemized INR dues, reviewer remarks, and HMAC-SHA256 digital seals.
- **`DS-5: Grievances & SLA Timers`** (**`grievances`** table): Disputed dues tickets, priority classes, computed SLA deadlines, and escalation flags.
- **`DS-6: System Audit Logs`** (**`audit_logs`** table): Append-only audit events tracking actors, actions, timestamps, and JSON-encoded metadata.

3.3 Level 1 System Decomposition Diagram (Mermaid Model)



3.4 Level 1 Inter-Process, Store, and Entity Data Flows

Process ID	Process Name	Consumed Inputs (From Entities & Stores)	Generated Outputs (To Entities & Stores)
`1.0`	**Authentication & Role Authorization**	`Student Credentials`, `Staff Credentials`, `Admin Credentials`, `DS-1: users` password hashes.	`JWT Access Tokens` (Student, Staff, Admin scopes with claims: `sub`, `role`, `exp`).
`2.0`	**Clearance Application & Initialization**	`Clearance Application` from Student, `DS-3: clearance_requests` (duplicate check), `DS-2: departments`.	Created record in `DS-3`, 5 initialized approval rows in `DS-4`, audit event to `6.0`, `Pipeline Status` to Student.
`3.0`	**Review, Chain of Responsibility & HMAC Signing**	`Due Inspection & Sign-off` from Staff, `Executive Override` from Admin, current approval states from `DS-4`.	Updated status & `HMAC Seal` in `DS-4`, unlocked downstream gates in `DS-4`, request status in `DS-3`, audit event to `6.0`.
`4.0`	**Grievance Redressal & SLA Escalation**	`Grievance Submission` from Student, `Resolution Decision` from Staff, `SLA Trigger` from Admin, deadline scans on `DS-5`.	Ticket recorded in `DS-5`, updated dues in `DS-4`, `SLA Breach Alerts` to Admin, resolution notes to Student, audit event to `6.0`.
`5.0`	**Certificate Minting & Verification**	Verified approvals from `DS-4`, cleared request from `DS-3`, `Verification Query` from Verifier.	`Cert ID` and `SHA-256 Digest` stored in `DS-3`, `Official PDF` streamed to Student, `Authenticity Record` to Verifier.
`6.0`	**Admin Analytics & Audit Logging**	Internal audit events from `2.0`, `3.0`, `4.0`, `5.0`; analytical aggregations across `DS-3`, `DS-4`, `DS-5`, `DS-6`.	Append-only records to `DS-6`, `Campus Analytics` & `Bottleneck Metrics` to Admin, `Audit Trail Logs` to Admin.

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4. Process Specifications (P-Specs), Decision Tables & Decision Trees

4.1 Process 3.0: Chain of Responsibility (CoR) Gating & Multi-Signatory Barrier Logic

Process Narrative & Algorithmic Specification:

Process `3.0` coordinates the multi-department approval workflow using the behavioral **Chain of Responsibility** pattern (`backend/app/patterns/chain_of_responsibility.py`). It enforces strict barrier rules:

1. **Gate 1 (Parallel Gates):** Dr. B.R. Ambedkar Central Library, PEC Hostels, and Gymkhana evaluate clearances concurrently. Neither blocks the other.
2. **Gate 2 (Sequential Dependency Barrier):** Departmental Laboratories (CSE/ECE/Mech/etc.) verify hardware and lab equipment returns **strictly after all Gate 1 facilities are APPROVED**. Any review attempt

before Gate 1 is 100% complete is rejected with `Prerequisite Gate Error`.

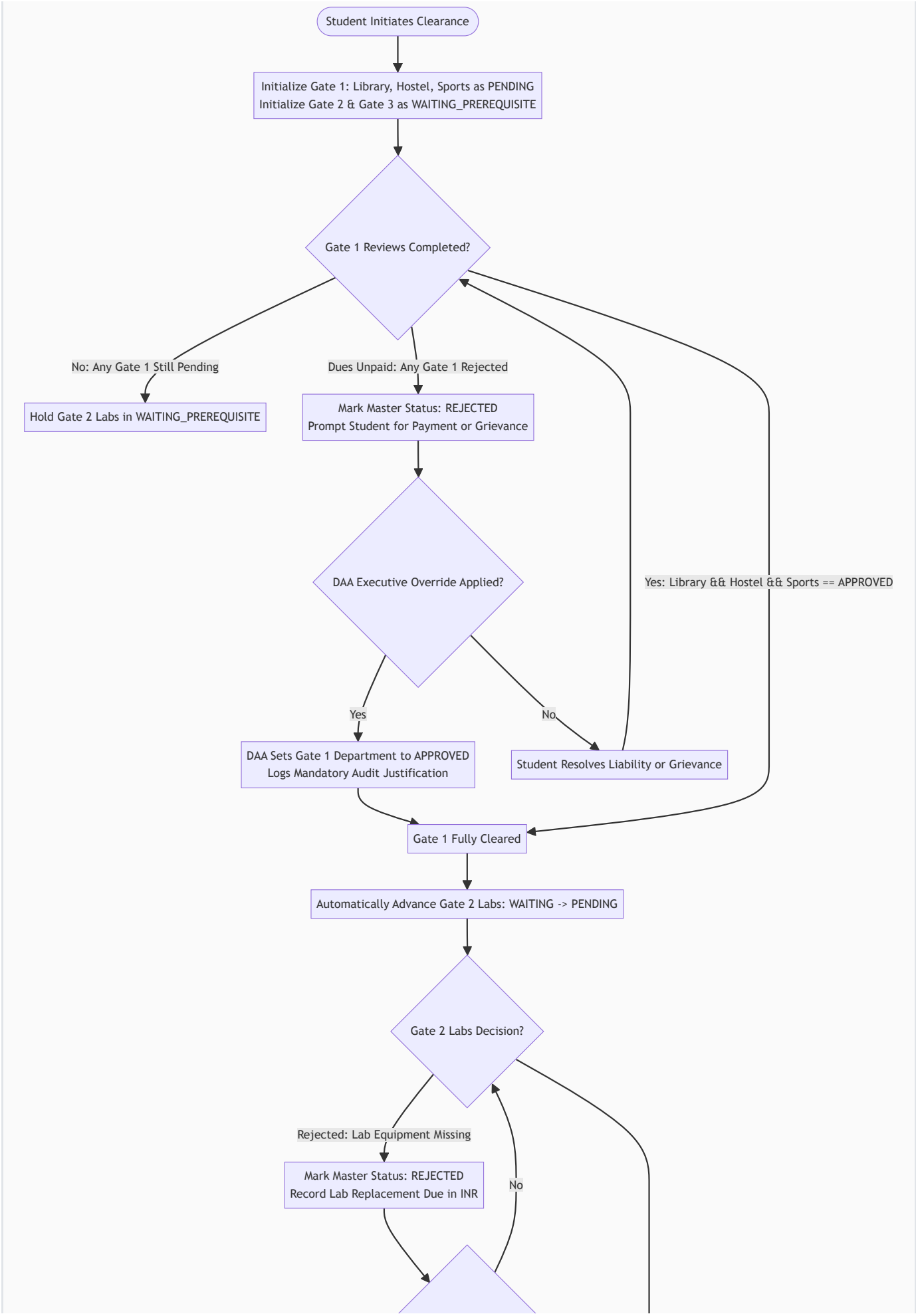
3. **Gate 3 (Final Gate Barrier):** Accounts Branch reconciles caution money and fee balances **strictly after Gates 1 and 2 are 100% APPROVED**.
4. **Cryptographic Sealing:** An approved review generates an irreversible HMAC-SHA256 signature seal: $$\text{Digital Signature} = \text{HMAC-SHA256}(K_{\text{secret}}, \text{staff_id} \parallel \text{dept_code} \parallel \text{request_id} \parallel \text{timestamp})$$
5. **Caution Money Disbursement:** Approval at Gate 3 unblocks the release authorization of the student's institutional Caution Money deposit (₹10,000.00).

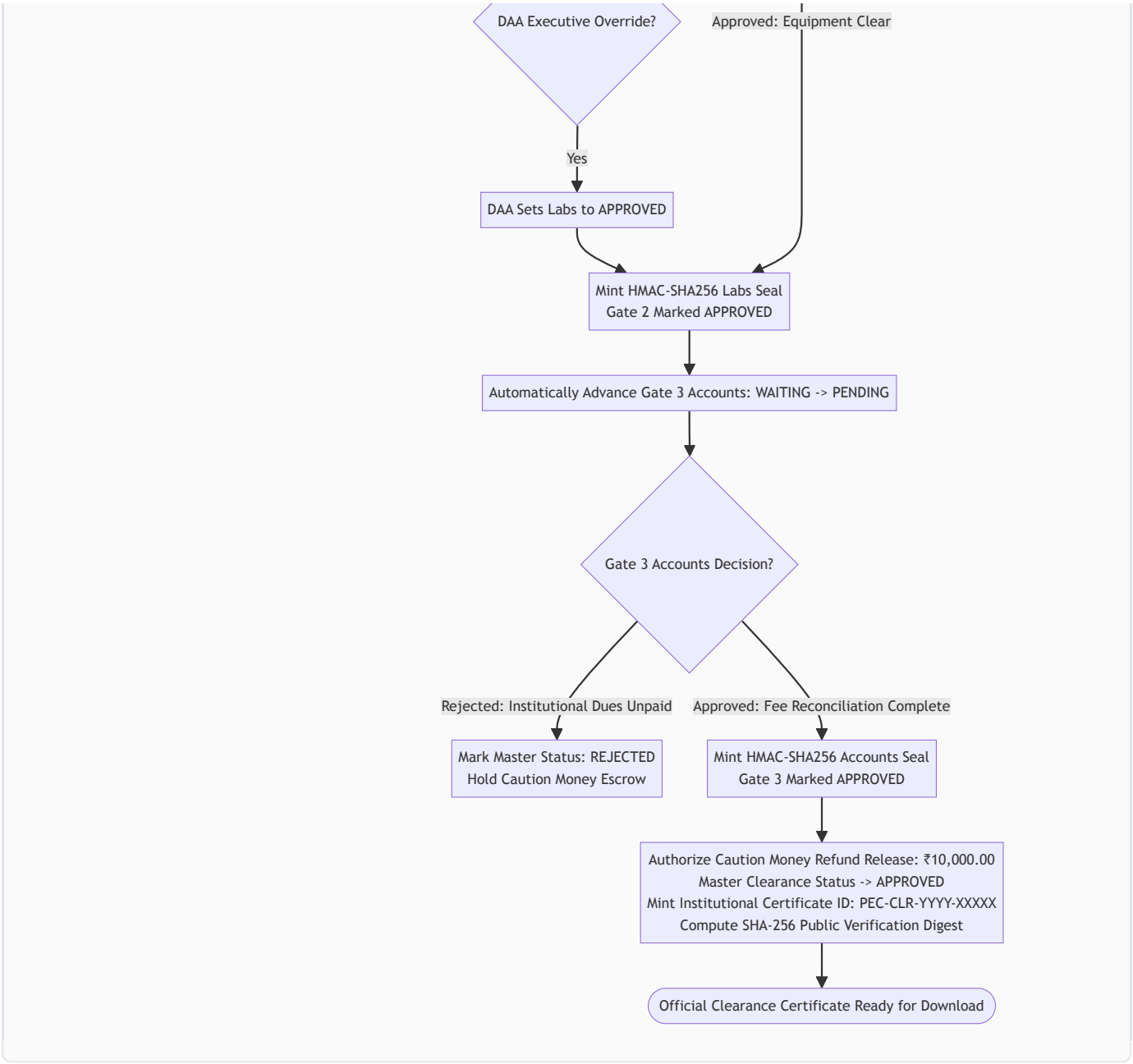
Decision Table 1: Multi-Gate Clearance Pipeline & Prerequisite Barrier Matrix

Rule ID	Gate 1 Library Status	Gate 1 Hostel Status	Gate 1 Sports Status	Gate 2 Labs Status	Gate 3 Accounts Status	DAA Override Active?	Department Target of Current Action
R-101	PENDING	PENDING	PENDING	WAITING	WAITING	No	Gate 1 (Lib/Host/Sport)
R-102	APPROVED	APPROVED	PENDING	WAITING	WAITING	No	Gate 2 (Dept Labs)
R-103	APPROVED	APPROVED	REJECTED	WAITING	WAITING	No	Gate 2 (Dept Labs)
R-104	APPROVED	APPROVED	APPROVED	WAITING	WAITING	No	System Evaluation
R-105	APPROVED	APPROVED	APPROVED	PENDING	WAITING	No	Gate 2 (Dept Labs)
R-106	APPROVED	APPROVED	APPROVED	APPROVED	WAITING	No	System Evaluation
R-107	APPROVED	APPROVED	APPROVED	APPROVED	PENDING	No	Gate 3 (Accounts)

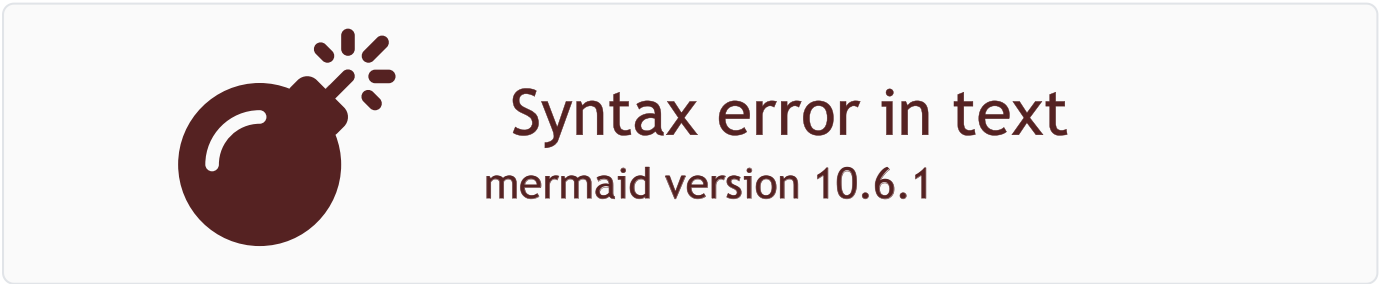
Rule ID	Gate 1 Library Status	Gate 1 Hostel Status	Gate 1 Sports Status	Gate 2 Labs Status	Gate 3 Accounts Status	DAA Override Active?	Department Target of Current Action
R-108	`APPROVED`	`APPROVED`	`APPROVED`	`APPROVED`	`APPROVED`	No	System Evaluation
R-109	Any	Any	`REJECTED`	Any	Any	**Yes (DAA)**	Gate 1 (Sports)

Decision Tree 1: Clearance Gating & Barrier Enforcement Tree (Mermaid)


Student Initiates Clearance
Initialize Gate 1: Library, Hostel, Sports as PENDING
Initialize Gate 2 & Gate 3 as WAITING_PREREQUISITE
Gate 1 Reviews Completed?
No: Any Gate 1 Still Pending
Hold Gate 2 Labs in WAITING_PREREQUISITE
Dues Unpaid: Any Gate 1 Rejected
Mark Master Status: REJECTED
Prompt Student for Payment or Grievance
DAA Executive Override Applied?
Yes
DAA Sets Gate 1 Department to APPROVED
Logs Mandatory Audit Justification
No
Student Resolves Liability or Grievance
Yes: Library && Hostel && Sports == APPROVED
Gate 1 Fully Cleared
Automatically Advance Gate 2 Labs: WAITING -> PENDING
Gate 2 Labs Decision?
Rejected: Lab Equipment Missing
Mark Master Status: REJECTED
Record Lab Replacement Due in INR
No



Decision Tree 2: Departmental Liability Assessment & HMAC Signing Flow (Mermaid)



4.2 Process 4.0: Grievance Redressal & SLA Escalation Engine

Process Narrative & Algorithmic Specification:

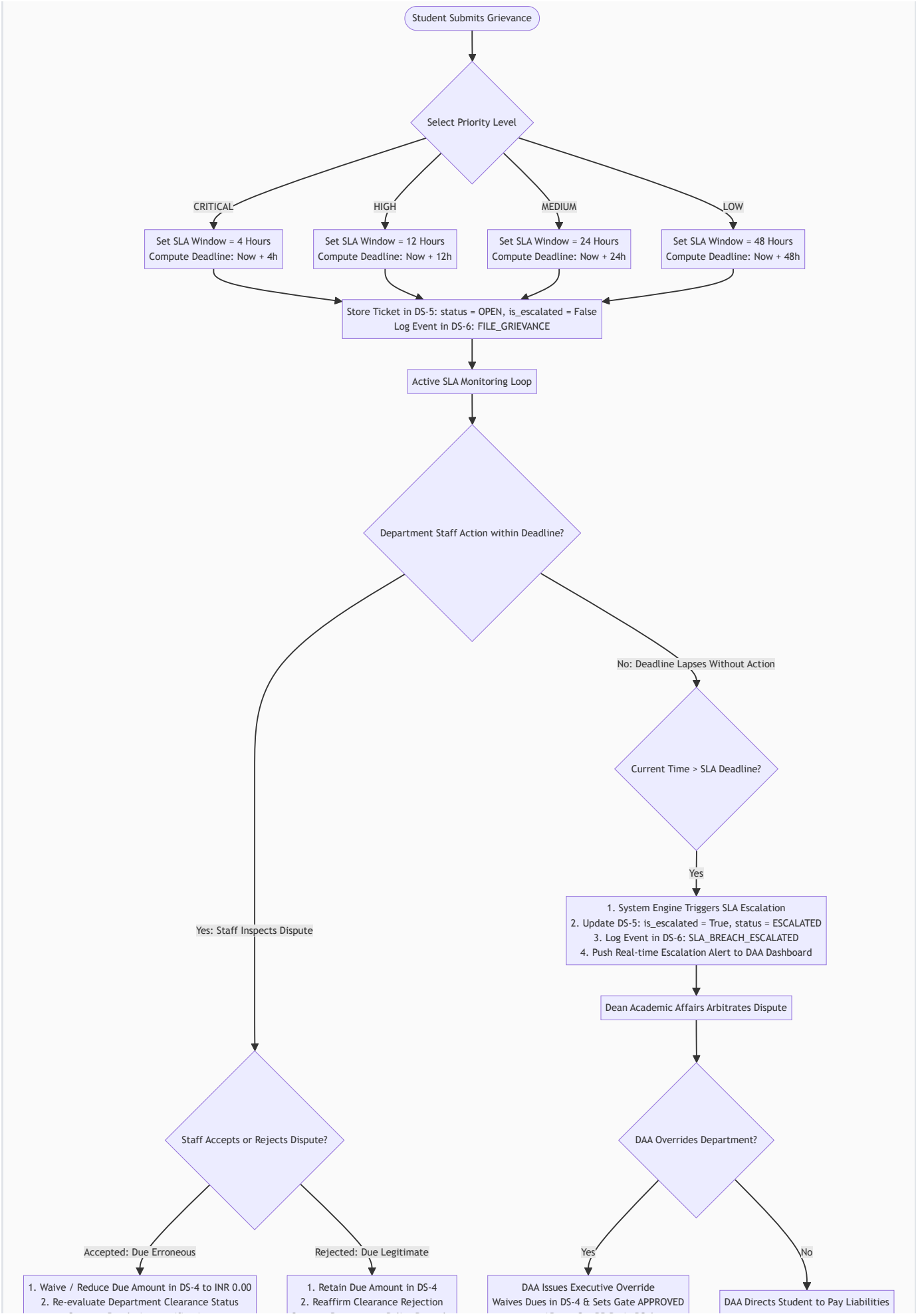
Process `4.0` governs student appeals regarding disputed fines (e.g., contested hostel mess bills or library replacement costs). It enforces dynamic Service Level Agreements (SLA):

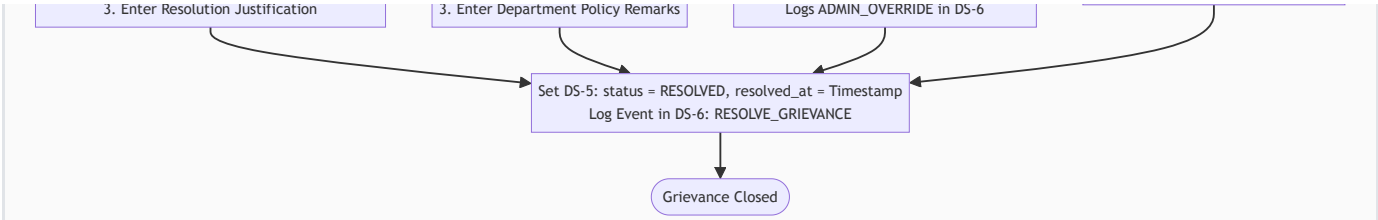
1. **Dynamic Deadline Calculation:** $$\text{SLA Deadline Timestamp} = \text{Submission Timestamp} + \text{SLA Window}(\text{Priority})$$
   - CRITICAL Priority: **4 Hours**
   - HIGH Priority: **12 Hours**
   - MEDIUM Priority: **24 Hours**
   - LOW Priority: **48 Hours**
2. **Real-time Escalation Scanner:** Evaluates all tickets where `Current_Time > sla_deadline` and `status in ('OPEN', 'IN_REVIEW')`. Expired tickets are updated to `status = 'ESCALATED'` with `is_escalated = True`.
3. **Executive Alerts:** Breached tickets generate immediate visual escalation banners on the Dean Academic Affairs dashboard.
4. **Dispute Settlement:** Resolution by staff or DAA updates the ticket to `RESOLVED`, sets `resolved_at`, adjusts or waives the disputed fine in `DS-4`, and logs the settlement in `DS-6`.

Decision Table 2: Priority SLA Deadline & Automated Escalation Matrix

Rule ID	Ticket Priority	Elapsed Time ($t_{\text{now}} - t_{\text{submitted}}$)	Current Ticket Status	Department Official Action	System SLA Engine Action	Maste Sta Mut
R-201	`CRITICAL`	$\le$ 4 Hours	`OPEN`	Pending review	Maintain live countdown timer (4h window).	No c (`OI` `is_esc` `= Fa`
R-202	`CRITICAL`	**> 4 Hours**	`OPEN`	None taken	**TRIGGER AUTO-ESCALATION:** Flag ticket for DAA; log breach event.	**`ESCA`** (`is_es` `= T`
R-203	`HIGH`	$\le$ 12 Hours	`OPEN`	Pending review	Maintain live countdown timer (12h window).	No c (`OI` `is_esc` `= Fa`
R-204	`HIGH`	**> 12 Hours**	`OPEN`	None taken	**TRIGGER AUTO-ESCALATION:** Flag ticket for DAA; log breach event.	**`ESCA`** (`is_es` `= T`
R-205	`MEDIUM`	$\le$ 24 Hours	`OPEN`	Pending review	Maintain live countdown timer (24h window).	No c (`OI` `is_esc` `= Fa`
R-206	`MEDIUM`	**> 24 Hours**	`OPEN`	None taken	**TRIGGER AUTO-ESCALATION:** Flag ticket for DAA; log breach event.	**`ESCA`** (`is_es` `= T`
R-207	`LOW`	$\le$ 48 Hours	`OPEN`	Pending review	Maintain live countdown timer (48h window).	No c (`OI` `is_esc` `= Fa`
R-208	`LOW`	**> 48 Hours**	`OPEN`	None taken	**TRIGGER AUTO-ESCALATION:** Flag ticket for DAA; log breach event.	**`ESCA`** (`is_es` `= T`
R-209	Any	Any	`OPEN` / `ESCALATED`	Staff/DAA registers resolution remarks	Record `staff_response`; zero or adjust INR dues in `DS-4`.	**`RES`** (`resol` `= N`

Decision Tree 3: Grievance Arbitration & SLA Breach Escalation Tree (Mermaid)


Student Submits Grievance
Select Priority Level
CRITICAL
HIGH
MEDIUM
LOW
Set SLA Window = 4 Hours
Compute Deadline: Now + 4h
Set SLA Window = 12 Hours
Compute Deadline: Now + 12h
Set SLA Window = 24 Hours
Compute Deadline: Now + 24h
Set SLA Window = 48 Hours
Compute Deadline: Now + 48h
Store Ticket in DS-5: status = OPEN, is_escalated = False
Log Event in DS-6: FILE_GRIEVANCE
Active SLA Monitoring Loop
Department Staff Action within Deadline?
Yes: Staff Inspects Dispute
No: Deadline Lapses Without Action
Current Time > SLA Deadline?
Yes
1. System Engine Triggers SLA Escalation
2. Update DS-5: is_escalated = True, status = ESCALATED
3. Log Event in DS-6: SLA_BREACH_ESCALATED
4. Push Real-time Escalation Alert to DAA Dashboard
Dean Academic Affairs Arbitrates Dispute
Staff Accepts or Rejects Dispute?
DAA Overrides Department?
Accepted: Due Erroneous
Rejected: Due Legitimate
Yes
No
1. Waive / Reduce Due Amount in DS-4 to INR 0.00
2. Re-evaluate Department Clearance Status
1. Retain Due Amount in DS-4
2. Reaffirm Clearance Rejection
DAA Issues Executive Override
Waives Dues in DS-4 & Sets Gate APPROVED
DAA Directs Student to Pay Liabilities



4.3 Process 1.0 & 3.0: Role-Based Access Control (RBAC) & Authority Validation

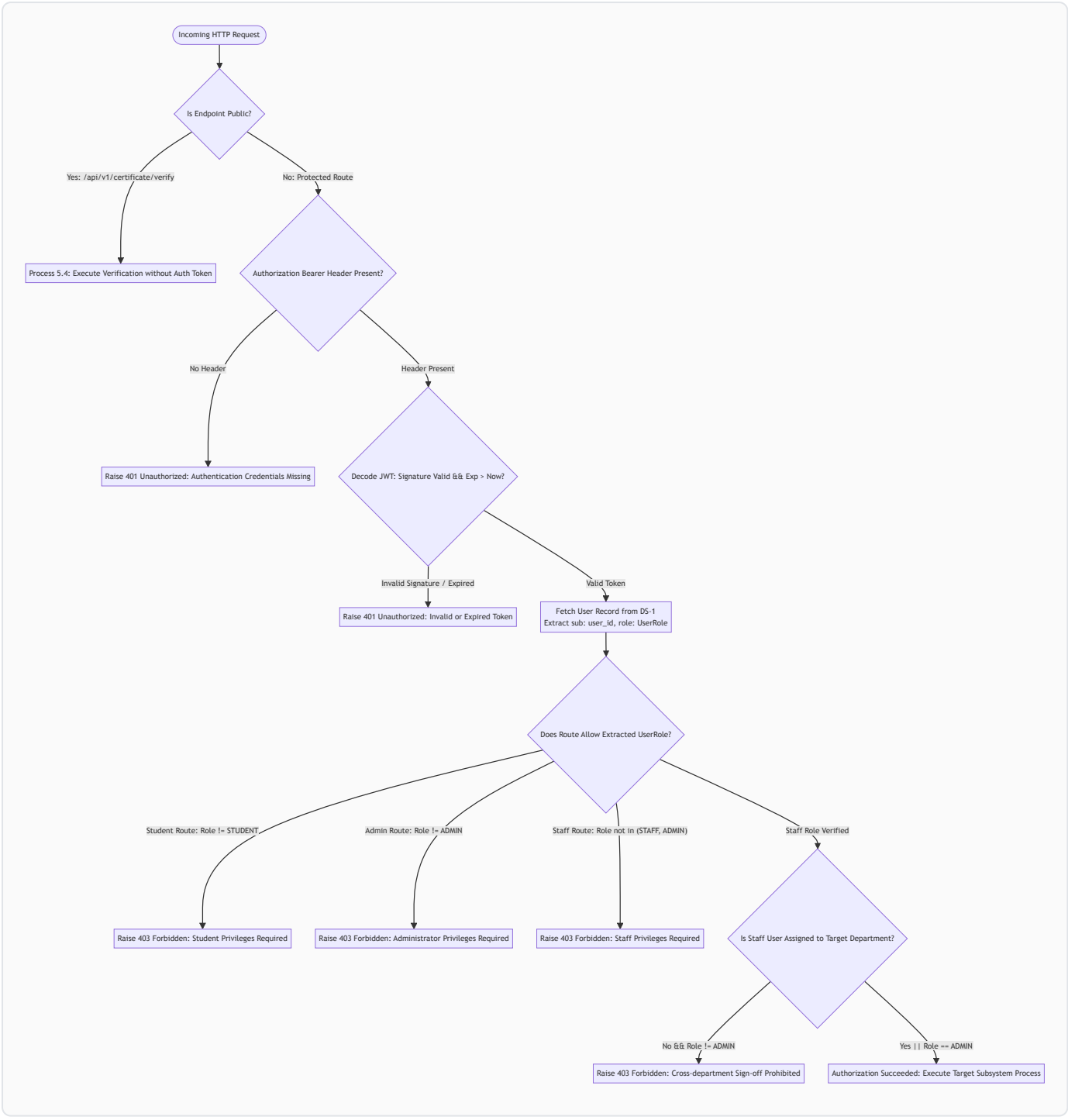
Process Narrative & Algorithmic Specification:

The API Gateway enforces strict **Role-Based Access Control (RBAC)** across four user scopes: `STUDENT`, `STAFF`, `ADMIN` (DAA), and unauthenticated `PUBLIC`. Unauthorized requests are blocked with HTTP `401 Unauthorized` or `403 Forbidden`.

Decision Table 3: Comprehensive RBAC Authorization Matrix

Action / Operation	API Route Endpoint	`STUDENT`	`STAFF` (Own Dept)	`STAFF` (Other Dept)	`ADMIN` (DAA)	
Apply for Clearance	`POST /api/v1/clearance/apply`	**ALLOW**	DENY	DENY	DENY	
View My Pipeline	`GET /api/v1/clearance/my-requests`	**ALLOW**	DENY	DENY	DENY	
View Staff Queue	`GET /api/v1/clearance/staff/queue`	DENY	**ALLOW**	DENY	**ALLOW**	
Sign Department Gate	`POST /api/v1/clearance/staff/decision/{id}`	DENY	**ALLOW**	DENY	**ALLOW**	
File Grievance Ticket	`POST /api/v1/grievance`	**ALLOW**	DENY	DENY	DENY	
Resolve Grievance	`POST /api/v1/grievance/{id}/resolve`	DENY	**ALLOW**	DENY	**ALLOW**	
Trigger SLA Scan	`POST /api/v1/grievance/trigger-sla-check`	DENY	**ALLOW**	**ALLOW**	**ALLOW**	
View Campus Analytics	`GET /api/v1/admin/analytics`	DENY	**ALLOW**	**ALLOW**	**ALLOW**	
View System Audit Logs	`GET /api/v1/admin/audit-logs`	DENY	DENY	DENY	**ALLOW**	
Execute Admin Override	`POST /api/v1/admin/clearance/{id}/override`	DENY	DENY	DENY	**ALLOW**	
Download PDF Cert	`GET /api/v1/certificate/download/{id}`	**ALLOW**	**ALLOW**	**ALLOW**	**ALLOW**	
Public Verification	`GET /api/v1/certificate/verify`	**ALLOW**	**ALLOW**	**ALLOW**	**ALLOW**	

Decision Tree 4: User Authentication & Authorization Gate Tree (Mermaid)



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5. Comprehensive Data Dictionary

5.1 Data Stores at Rest (`DS-1` to `DS-6` Schemas)

The schemas reflect the exact production relational database definitions (`backend/app/models/models.py`):

`DS-1` : Users & Personas (`users` table)

Column Name	SQL Type	Constraint	Description	Concrete Example
`id`	`INTEGER`	Primary Key	Auto-incrementing surrogate identifier	`1`
`email`	`VARCHAR(120)`	Unique, Not Null	Institutional PEC email address	`alice@pec.edu.in`
`full_name`	`VARCHAR(120)`	Not Null	User full name or administrative title	`Alice Sharma` / `Chief Warden`
`role`	`ENUM`	Not Null	Security role: `STUDENT`, `STAFF`, `ADMIN`	`STUDENT`
`roll_number`	`VARCHAR(50)`	Unique, Nullable	8-digit PEC SID format (`YYBBBRRR`)	`22103042`
`department_id`	`INTEGER`	Foreign Key (`departments.id`)	Foreign key linking staff to their facility	`1` (Library)
`hashed_password`	`VARCHAR(255)`	Not Null	Salted Bcrypt hash (work factor 12)	`$2b$12$e8x...`
`created_at`	`DATETIME`	Default UTC	Timestamp of user provisioning	`2026-05-01 09:00:00`

`DS-2`: Campus Departments (`departments` table)

Column Name	SQL Type	Constraint	Description	Concrete Example
`id`	`INTEGER`	Primary Key	Department surrogate identifier	`1`
`name`	`VARCHAR(100)`	Not Null	Official campus facility designation	`Dr. B. R. Ambedkar Central Library`
`code`	`VARCHAR(50)`	Unique, Not Null	Institutional department code	`LIBRARY`, `HOSTEL`, `SPORTS`, `LABS`, `ACCOUNTS`
`clearance_gate`	`INTEGER`	Not Null	Pipeline gate level: `1` (Parallel), `2` (Labs), `3` (Accounts)	`1`
`description`	`VARCHAR(255)`	Nullable	Detailed functional remit	`Central university library textbook dues`
`staff_head_name`	`VARCHAR(100)`	Nullable	Positional title of head signatory	`Chief Librarian`

`DS-3`: Clearance Requests (`clearance_requests` table)

Column Name	SQL Type	Constraint	Description	Concrete Example
`id`	`INTEGER`	Primary Key	Clearance request unique identifier	`42`
`student_id`	`INTEGER`	Foreign Key (`users.id`)	References applicant student	`1` (Alice Sharma)
`academic_year`	`VARCHAR(20)`	Not Null	Enrolled academic year session	`2025–2026`
`semester`	`VARCHAR(20)`	Not Null	Current academic semester	`Semester 8`
`reason`	`VARCHAR(100)`	Not Null	Institutional justification	`Graduation Clearance & Degree Awa`
`overall_status`	`ENUM`	Not Null	Master status: `PENDING` , `IN_PROGRESS` , `APPROVED` , `REJECTED`	`APPROVED`
`certificate_id`	`VARCHAR(64)`	Unique, Nullable	Minted certificate number	`PEC-CLR-2026-00042`
`certificate_hash`	`VARCHAR(128)`	Nullable	SHA-256 public verification digest	`7f83b1657ff1fc53b92dc18148a1d65d`
`created_at`	`DATETIME`	Default UTC	Timestamp of clearance application	`2026-05-01 10:00:00`
`updated_at`	`DATETIME`	On Update UTC	Timestamp of last state mutation	`2026-05-03 14:30:00`

`DS-4` : Department Approvals (**`department_approvals`** table)

Column Name	SQL Type	Constraint	Description	Concrete E
`id`	`INTEGER`	Primary Key	Approval record surrogate key	`201`
`request_id`	`INTEGER`	Foreign Key (`clearance_requests.id`)	References parent clearance request	`42`
`department_id`	`INTEGER`	Foreign Key (`departments.id`)	References reviewing department	`1` (Library
`status`	`ENUM`	Not Null	`WAITING_PREREQUISITE` , `PENDING` , `APPROVED` , `REJECTED`	`APPROVED`
`staff_id`	`INTEGER`	Foreign Key (`users.id`)	References signing staff official	`2` (Librari
`approved_at`	`DATETIME`	Nullable	Exact timestamp of review sign-off	`2026-05-0`
`remarks`	`TEXT`	Nullable	Itemized inspection commentary	`All libra for. Zero`
`due_amount`	`FLOAT`	Default 0.0	Outstanding liability in INR (`₹`)	`0.00`
`digital_signature`	`VARCHAR(255)`	Nullable	Keyed HMAC-SHA256 signature seal	`SIG-v1:LIBRARY`

`DS-5`: Grievances & SLA Timers (`grievances` table)

Column Name	SQL Type	Constraint	Description	Concrete Example
`id`	`INTEGER`	Primary Key	Grievance ticket identifier	`8`
`student_id`	`INTEGER`	Foreign Key (`users.id`)	References appellant student	`3` (Charlie Verma)
`clearance_request_id`	`INTEGER`	Foreign Key (`clearance_requests.id`)	Associated clearance application	`42`
`department_id`	`INTEGER`	Foreign Key (`departments.id`)	Target disputed department	`2` (Hostels)
`subject`	`VARCHAR(200)`	Not Null	Summary dispute title	`Shivalik Mess Dues Inaccuracy`
`description`	`TEXT`	Not Null	Detailed appeal justification	`Internship sanctioned by DAA for Jan–June`
`priority`	`ENUM`	Not Null	`LOW` , `MEDIUM` , `HIGH` , `CRITICAL`	`HIGH`
`status`	`ENUM`	Not Null	`OPEN` , `IN_REVIEW` , `RESOLVED` , `ESCALATED`	`ESCALATED`
`sla_hours`	`INTEGER`	Default 24	Allowed resolution window in hours	`12`
`sla_deadline`	`DATETIME`	Not Null	Computed absolute expiration deadline	`2026-05-02 22:00:00`
`is_escalated`	`BOOLEAN`	Default False	True if deadline passed without resolution	`True`

Column Name	SQL Type	Constraint	Description	Concrete Example
`staff_response`	`TEXT`	Nullable	Final written response from authority	`Mess rebate verified; dues waived.`
`created_at`	`DATETIME`	Default UTC	Submission timestamp	`2026-05-02 10:00:00`
`resolved_at`	`DATETIME`	Nullable	Exact timestamp of resolution	`2026-05-03 08:30:00`

`DS-6` : System Audit Logs (`audit_logs` table)

Column Name	SQL Type	Constraint	Description	Concrete Example
`id`	`INTEGER`	Primary Key	Sequential audit entry index	`1052`
`actor_id`	`INTEGER`	Nullable	User ID of the mutating actor	`2`
`actor_name`	`VARCHAR(120)`	Not Null	Human-readable name of the actor	`Chief Librarian` / `SLA_ENGINE`
`actor_role`	`VARCHAR(50)`	Not Null	Security role: `STUDENT`, `STAFF`, `ADMIN`, `SYSTEM`	`STAFF`
`action`	`VARCHAR(100)`	Not Null	Immutable action verb	`LIBRARY_APPROVED`, `ADMIN_OVERRIDE`
`target_type`	`VARCHAR(50)`	Not Null	Entity category: `CLEARANCE`, `GRIEVANCE`	`CLEARANCE`
`target_id`	`INTEGER`	Nullable	Foreign ID of the affected entity	`42`
`details`	`TEXT`	Nullable	Textual metadata and mutation payload	`Decision: APPROVED. Dues: ₹0.00. Sig: SIG-v1...`
`timestamp`	`DATETIME`	Default UTC	Microsecond precision timestamp	`2026-05-02 11:15:02`

5.2 Data Elements in Motion (Composite Packets & Primitive Elements)

Composite Data Packets:

1. `Clearance_Submission_Packet`: $$\text{Packet} = {\text{student_sid: String}, \text{academic_year: String}, \text{semester: String}, \text{reason: String}}$$
2. `Department_Signoff_Packet`: $$\text{Packet} = {\text{approval_id: Int}, \text{decision: Enum}, \text{remarks: String}, \text{due_amount: Float} \ge 0.00}$$
3. `Grievance_Filing_Packet`: $$\text{Packet} = {\text{department_id: Int}, \text{subject: String}, \text{description: String}, \text{priority: Enum}}$$
4. `Public_Verification_Result_Packet`: $$\text{Packet} = {\text{valid: Bool}, \text{cert_id: String}, \text{student_name: String}, \text{sid: String}, \text{signatures: Array[HMAC_Seal]}}$$

Primitive Data Elements:

- `SID` **(Student Identification Number):**
  - **Type:** String (8 digits).
  - **Regular Expression:** `^[0-9]{8}$` (e.g., `22103042`).
  - **Domain:** Valid Punjab Engineering College student enrollment codes.
- `due_amount` **(Departmental Liability):**
  - **Type:** Float (2 decimal precision).
  - **Currency:** Indian Rupee (`₹` / INR).
  - **Domain:** `[0.00, 500000.00]`. Negative numbers are rejected by schema validators with `422 Unprocessable Entity`.
- `digital_signature` **(HMAC Approval Seal):**
  - **Type:** String (ASCII, formatted).
  - **Syntax:** `SIG-v1:<DEPT_CODE>:<24_CHAR_HEX>` (e.g., `SIG-v1:LIBRARY:a7f9b2c3d4e5f6018293a4b5`).
  - **Security Property:** Non-repudiable; invalidates if staff ID, department, request ID, or timestamp are altered.
- `certificate_hash` **(Public Verification Digest):**
  - **Type:** String (64-character lowercase hexadecimal).
  - **Formula:** `SHA-256(cert_id + ":" + student_id + ":" + academic_year + ":" + semester)`

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6. Formal Horizontal & Vertical Balancing Verification

To ensure that the DFD specifications contain **zero structural discrepancies** and are **frozen permanently**, the table below verifies horizontal and vertical parity between Level 0 and Level 1:

Level 0 Boundary Flow Index	Boundary Flow Name	External Entity	Level 0 Direction	Level 1 Connecting Subsystem Process	Balancing Parity Status
F-01	`Clearance Application`	`Student`	Input	Process `2.0: Clearance Application & Workflow`	BALANCED (100%)
F-02	`Grievance Submission`	`Student`	Input	Process `4.0: Grievance Redressal & SLA Engine`	BALANCED (100%)
F-03	`Student Auth Credentials`	`Student`	Input	Process `1.0: Authentication & Role Authorization`	BALANCED (100%)
F-04	`Auth Token & Session Scope`	`Student`	Output	Process `1.0: Authentication & Role Authorization`	BALANCED (100%)
F-05	`Live Pipeline Status`	`Student`	Output	Process `2.0: Clearance Application & Workflow`	BALANCED (100%)
F-06	`Grievance Resolution Updates`	`Student`	Output	Process `4.0: Grievance Redressal & SLA Engine`	BALANCED (100%)
F-07	`Official PDF Certificate`	`Student`	Output	Process `5.0: Certificate Minting & Verification`	BALANCED (100%)
F-08	`Staff Auth Credentials`	`Staff`	Input	Process `1.0: Authentication & Role Authorization`	BALANCED (100%)
F-09	`Due Inspection & Sign-off`	`Staff`	Input	Process `3.0: Review, Chain of Responsibility & HMAC`	BALANCED (100%)
F-10	`Grievance Resolution Decision`	`Staff`	Input	Process `4.0: Grievance Redressal & SLA Engine`	BALANCED (100%)
F-11	`Staff Auth Token`	`Staff`	Output	Process `1.0: Authentication & Role Authorization`	BALANCED (100%)

Level 0 Boundary Flow Index	Boundary Flow Name	External Entity	Level 0 Direction	Level 1 Connecting Subsystem Process	Balancing Parity Status
F-12	`Department Review Queue`	`Staff`	Output	Process `3.0: Review, Chain of Responsibility & HMAC`	BALANCED (100%)
F-13	`Department Grievance Worklist`	`Staff`	Output	Process `4.0: Grievance Redressal & SLA Engine`	BALANCED (100%)
F-14	`Admin Auth Credentials`	`Admin`	Input	Process `1.0: Authentication & Role Authorization`	BALANCED (100%)
F-15	`Executive Override Command`	`Admin`	Input	Process `3.0: Review, Chain of Responsibility & HMAC`	BALANCED (100%)
F-16	`SLA Evaluation Trigger`	`Admin`	Input	Process `4.0: Grievance Redressal & SLA Engine`	BALANCED (100%)
F-17	`Admin Auth Token`	`Admin`	Output	Process `1.0: Authentication & Role Authorization`	BALANCED (100%)
F-18	`Campus Clearance Analytics`	`Admin`	Output	Process `6.0: Administration, Analytics & Logging`	BALANCED (100%)
F-19	`Escalated Grievance Alerts`	`Admin`	Output	Process `4.0: Grievance Redressal & SLA Engine`	BALANCED (100%)
F-20	`System Audit Trail Logs`	`Admin`	Output	Process `6.0: Administration, Analytics & Logging`	BALANCED (100%)
F-21	`Public Verification Query`	`Verifier`	Input	Process `5.0: Certificate Minting & Verification`	BALANCED (100%)
F-22	`Authenticity Record & Seals`	`Verifier`	Output	Process `5.0: Certificate Minting & Verification`	BALANCED (100%)

Conclusion: All 22 boundary data flows exhibit exact horizontal and vertical parity between Level 0 and Level 1. The data stores (`DS-1` through `DS-6`) match production relational tables, decision tables cover all operational branching conditions, decision trees visually articulate complex state lifecycles, and no lower-level

decompositions exceed Level 1. This document constitutes the **Frozen Final Baseline** of the Structured Systems Analysis for the Punjab Engineering College Clearance Automation Engine.